

DevOps Project

1. Project Title

Online Learning Platform (OLP)

2. Project Overview

The **Online Learning Portal (OLP)** is a web-based application designed to streamline and manage academic activities for students, faculty, parents, and administrators. The system provides features such as student attendance tracking, assignment management, result viewing, course management, and real-time notifications through a centralized platform. It enables faculty to manage academic records efficiently while allowing parents to monitor their child's academic progress. The portal improves communication, reduces manual paperwork, and enhances the overall learning experience. By integrating DevOps practices such as Git, GitHub, Jenkins, Docker, Kubernetes, Prometheus, and Grafana, the system ensures continuous integration, automated deployment, scalability, and reliable performance.

3. Problem Statement

Many educational institutions still rely on manual processes or multiple disconnected systems to manage attendance, assignments, academic results, and student progress. This often leads to data inconsistency, communication gaps, and increased administrative effort. The Online Learning Portal addresses these challenges by providing a centralized platform for efficient academic management and real-time access to information.

4. Project Objectives

Project Objectives

1. Develop a centralized web-based platform for managing students, faculty, parents, and academic activities.
2. Automate attendance tracking, assignment submission, and result management to reduce manual work.
3. Provide secure role-based access for students, faculty, parents, and administrators.
4. Enable real-time notifications and academic progress monitoring for better communication.

5. Implement DevOps practices such as CI/CD, containerization, and cloud deployment to ensure scalability and reliability.

5. Users of the System

Admin

- Manages the entire Online Learning Portal.
- Adds, updates, and removes student, faculty, and parent records.
- Generates academic reports and analytics.
- Monitors attendance, assignments, results, and overall system activities.

Faculty

- Logs in to the portal securely.
- Manages student attendance and academic records.
- Uploads assignments, study materials, and marks.
- Monitors student performance and provides academic guidance.

Student

- Registers and maintains personal and academic information.
- Views attendance, assignments, course details, and results.
- Submits assignments and tracks submission status.
- Receives notifications and monitors academic progress.

Parent

- Logs in to the portal to monitor their child's academic activities.
- Views attendance records and academic performance.
- Tracks assignment completion and examination results.
- Receives notifications regarding student progress and important updates.

6. Functional Requirements

1. Home Page

- Project Overview
- Features of the System
- About Us

- Contact Us
- Login Button
- Register Button

Purpose:

The Home Page introduces the Online Learning Portal to users and provides access to the system.

How it Works:

- Displays information about the portal and its features.
- Provides Login and Registration options for users.
- Shows contact and support information.
- Allows users to navigate to the Registration or Login page.

2. Registration Page

- Student Name
- Student ID
- Email
- Course
- Semester
- Password
- Confirm Password
- Register Button

Purpose:

Allows new students, faculty members, and parents to create an account.

How it Works:

- User enters personal and academic details.
- System validates the information entered.
- User account is created and stored in the database.
- User is redirected to the Login Page.

3. Login Page

- Username/Email
- Password
- Login Button
- Forgot Password Link

Purpose:

Authenticates users before accessing the portal.

How it Works:

- User enters registered credentials.
- System verifies the credentials.
- If valid, the user is redirected to the Dashboard.
- If invalid, an error message is displayed.

4. Forgot Password Page

- Enter Email
- OTP Verification
- New Password
- Confirm Password
- Reset Password

Purpose:

Helps users recover their account password.

How it Works:

- User enters the registered email address.
- System sends an OTP or verification link.
- User verifies identity and creates a new password.
- Password is updated in the database.

5. Student Management Page

- Add Student
- Update Student Details
- View Student List
- Student Status
- Delete Student

Purpose:

Manages student information and academic records.

How it Works:

- Admin adds new student records.
- Admin updates student information.
- Admin searches and views student records.
- Students can view their updated information.
- Data is stored and maintained in the database.

6. Dashboard Page

- Total Students
- Total Faculty
- Total Courses
- Total Assignments
- Quick Statistics Cards

Purpose:

Provides an overview of academic activities and portal usage.

How it Works:

- Displays total students and faculty members.
- Shows attendance statistics.
- Displays pending assignments and notifications.
- Provides quick academic performance insights.

7. Faculty Management Page

- Add Faculty
- Edit Faculty
- Delete Faculty
- Faculty Details
- Faculty Availability

Purpose:

Manages faculty information and teaching activities.

How it Works:

- Admin adds faculty records.
- Admin updates faculty information.
- Admin removes inactive faculty records.
- System stores faculty details and teaching assignments.

8. Attendance Management Page

- Mark Attendance
- View Attendance
- Attendance Percentage
- Attendance Reports

Purpose:

Tracks and manages student attendance.

How it Works:

- Faculty records student attendance.
- System calculates attendance percentages automatically.
- Students and parents can view attendance records.
- Attendance reports are generated for analysis.

9. Assignment Management Page

- Upload Assignment
- Submit Assignment
- Assignment Status
- Due Date
- Assignment Reports

Purpose:

Manages assignment creation, submission, and tracking.

How it Works:

- Faculty uploads assignments.
- Students submit assignments before deadlines.
- System tracks submission status.
- Assignment history is stored for future reference.

10. Results Management Page

- Marks Entry
- Semester Results
- GPA Calculation
- Academic Performance Reports

Purpose:

Stores and manages student examination results.

How it Works:

- Faculty enters student marks.
- System calculates GPA and performance metrics.
- Students and parents can view results.
- Result records are stored securely in the database.

11. Reports and Notifications Page

- Student Report
- Attendance Report
- Assignment Report
- Result Report
- Download Reports

Purpose:

Generates academic reports and notifications.

How it Works:

- Generates attendance reports.
- Generates assignment and result reports.
- Sends notifications regarding academic activities.
- Reports can be downloaded as PDF or Excel files.

7. Non-Functional Requirements

8. Expected Benefits

- Reduced manual effort
- Improved healthcare tracking
- Better inventory control
- Faster recovery process
- Enhanced reporting and analysis

9. Product Backlog

Backlog ID	User Story / Requirement	Owner	Priority	Sprint
PB-01	Gather requirements from students, faculty, parents and stakeholders	Product Owner	High	Sprint-1
PB-02	Create Software Requirement Specification (SRS)	Product Owner	High	Sprint-1
PB-03	Design system architecture for Online Learning Portal	DevOps Engineer	High	Sprint-1
PB-04	Create database schema for students, faculty, attendance and results	Backend Developer	High	Sprint-1
PB-05	Setup GitHub repository and branching strategy	DevOps Engineer	High	Sprint-1
PB-06	Install and configure Jenkins for CI/CD	DevOps Engineer	High	Sprint-1

PB-07	As a student, I want to register and login securely	Frontend & Backend Developer	High	Sprint-2
PB-08	As a faculty member, I want to login securely	Frontend & Backend Developer	High	Sprint-2
PB-09	As a parent, I want to login securely	Frontend & Backend Developer	High	Sprint-2
PB-10	Implement role-based access control	Backend Developer	High	Sprint-2
PB-11	As a student, I want to view my dashboard	Frontend Developer	Medium	Sprint-2
PB-12	As a student, I want to view attendance details	Frontend & Backend Developer	High	Sprint-3
PB-13	As a faculty member, I want to enter attendance	Frontend & Backend Developer	High	Sprint-3
PB-14	As a student, I want to view assignments	Frontend & Backend Developer	High	Sprint-3
PB-15	As a faculty member, I want to upload assignments	Frontend & Backend Developer	Medium	Sprint-3
PB-16	As a student, I want to view semester results	Frontend & Backend Developer	High	Sprint-3
PB-17	As a parent, I want to monitor student progress	Frontend & Backend Developer	Medium	Sprint-3
PB-18	Implement notification module	Backend Developer	Medium	Sprint-3
PB-19	Create automated test cases	QA Engineer	High	Sprint-4
PB-20	Perform functional and API testing	QA Engineer	High	Sprint-4
PB-21	Containerize application using Docker	DevOps Engineer	High	Sprint-4
PB-22	Build Jenkins CI/CD pipeline	DevOps Engineer	High	Sprint-4
PB-23	Deploy application to cloud infrastructure	Cloud & Release Engineer	High	Sprint-4
PB-24	Configure Prometheus and Grafana monitoring	Cloud & Release Engineer	Medium	Sprint-4
PB-25	Prepare project documentation and sprint reports	Scrum Master	Medium	Sprint-4

10.Sprint Backlogs

Sprint 1: Requirement Analysis & Environment Setup

- Requirement Gathering
- User Stories Creation
- Architecture Design
- Database Design
- GitHub Repository Setup
- Jenkins Installation and Configuration
- Initial Testing
- Deliverable
- Product Backlog and Architecture Document

Sprint 2: Authentication & User Management

- Student Login Module
- Faculty Login Module
- Parent Login Module
- User Registration Module
- Role-Based Access Control
- Password Recovery Functionality
- Testing
- Deliverable
- Working Authentication and User Management System

Sprint 3: Academic Management

- Attendance Management Module
- Assignment Management Module
- Assignment Submission System
- Results Management Module
- Course Management Module
- Student, Faculty, and Parent Dashboards
- Notifications System
- Testing
- Deliverable
- Complete Academic Management System

Sprint 4: DevOps & Deployment

- Docker Containerization
- Jenkins CI/CD Pipeline Setup

- Automated Testing Integration
- Cloud Deployment (AWS/Azure)
- Prometheus Monitoring Setup
- Grafana Dashboard Configuration
- Bug Fixes and Final Testing
- Documentation Preparation
- Deliverable
- Production-Ready Online Learning Portal with Monitoring Dashboard

11. Production-ready application.

Admin

- As an Admin, I want to manage students, faculty, and parent accounts so that I can control system access.
- As an Admin, I want to generate academic reports so that I can analyze student performance and attendance.
- As an Admin, I want to manage courses so that academic information remains organized.

Student

- As a Student, I want to view my attendance online so that I can track my academic progress.
- As a Student, I want to submit assignments online so that I can complete coursework efficiently.
- As a Student, I want to view my results so that I can monitor my academic performance.
- As a Student, I want to receive notifications so that I stay informed about academic activities.

Faculty

- As a Faculty member, I want to record attendance so that student participation can be tracked.
- As a Faculty member, I want to upload assignments so that students can access and complete them.
- As a Faculty member, I want to enter marks and results so that student performance can be evaluated.

- As a Faculty member, I want to view student records so that I can provide academic guidance.

Parent

- As a Parent, I want to view my child's attendance so that I can monitor academic engagement.
- As a Parent, I want to view assignment status so that I can track academic responsibilities.
- As a Parent, I want to access results and performance reports so that I can support my child's learning progress.

DevOps Engineer

- As a DevOps Engineer, I want to automate application deployment so that software releases are faster and more reliable.
- As a DevOps Engineer, I want to monitor application performance so that issues can be detected and resolved quickly.

QA/Test Engineer

- As a QA Engineer, I want to test all modules so that the system remains reliable and free from defects.
- As a QA Engineer, I want to track and report bugs so that development teams can resolve issues efficiently.

11. Project Workflow

Home → Registration → Login → Dashboard → Student Management → Faculty Management → Course Management → Attendance Management → Assignment Management → Assignment Submission → Results Management → Notifications → Reports → Profile → Logout

10. Conclusion

The Online Learning Portal is a web-based application developed to simplify and automate academic activities such as attendance management, assignment handling, result management, and course administration. The system provides a centralized platform that improves communication among students, faculty, parents, and administrators while reducing manual effort and errors. By integrating DevOps practices, the portal ensures secure, scalable, reliable, and efficient delivery of educational services.